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Regional Centre for Space Science and Technology Education in Asia and the Pacific

Report

Spacecraft Structure Analysis and Design Experiment

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Abstract

This report focuses on the structural modeling, analysis, and optimization of a reinforced cylindrical shell under non-uniform internal pressure. The baseline material, copper, is replaced with Aluminum 6061-T6, and the composite layup is optimized from [±45°] to [90/0/Core/0/90], achieving substantial improvements in structural performance. The optimized design results in a **69% minimizing of the total mass**, a **31.63% reduction in maximum displacement**, and a reduction of the **cost** by approximately **60%**. Using **Classical Lamination Theory (CLT)** and **finite element modeling (FEA)** in MSC.Patran/Nastran, the results confirm the efficiency and structural integrity of the optimized design.

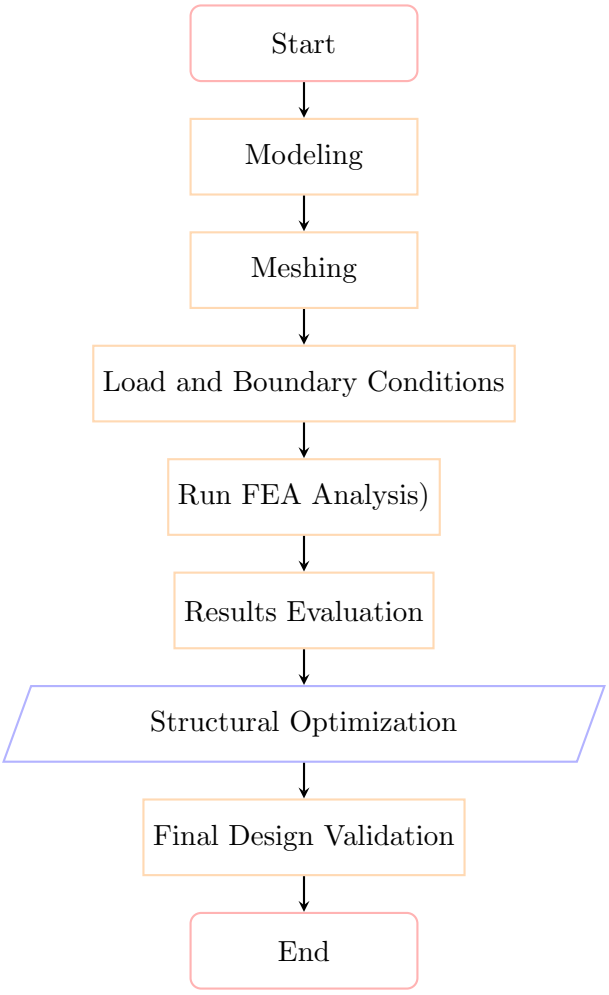


Figure 1: Flowchart of Optimization Process

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1. Structural Modeling Process

1.1 Detail of Structure

The radius of the cylindrical shell is $R = 0.5\,\text{m}$, and the length is $L = 6\,\text{m}$. The cylindrical shell is reinforced with longitudinal and circular beams made of metal. There are 8 longitudinal reinforced beams symmetrically distributed along the shell, with an L-shaped cross-section ($w = h = 10\,\text{mm}$, $t = 3\,\text{mm}$). Three circular reinforced beams with rectangular cross-sections ($w = h = 10\,\text{mm}$) are located at both ends ($Z = 0$, $Z = 6\,\text{m}$) and the middle ($Z = 3\,\text{m}$) of the structure.

The shell comprises two distinct sections: a composite section with thickness 6.2 mm from $Z = 0$ to $Z = 3\,\text{m}$ (fixed end to middle), and a metallic (copper) section with thickness 2 mm from $Z = 3\,\text{m}$ to $Z = 6\,\text{m}$ (middle to cantilever end).

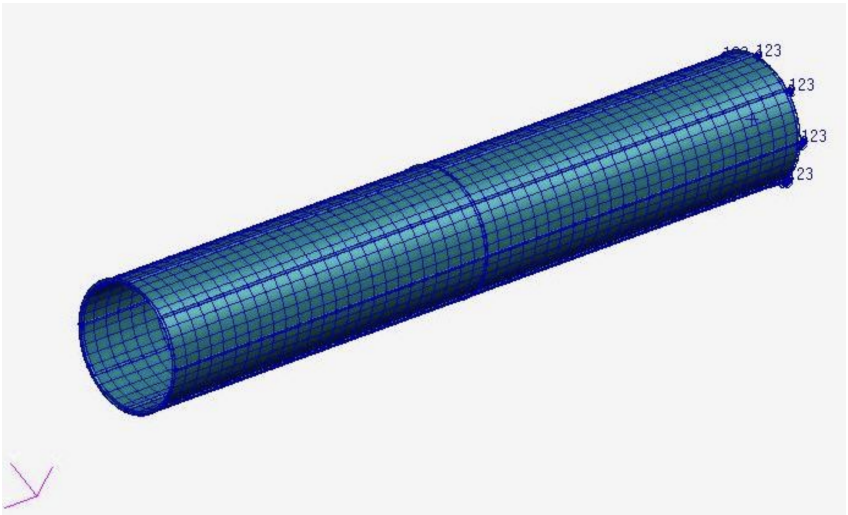


Figure 2: Cylindrical shell structure and its constraint.

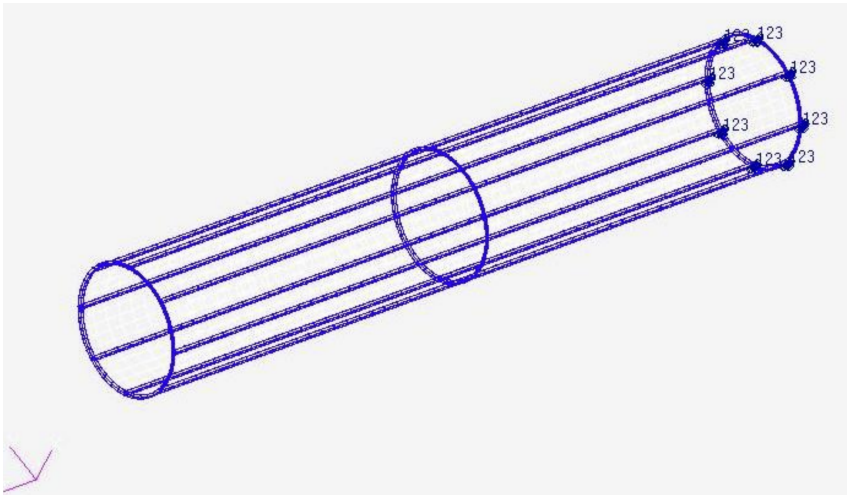


Figure 3: Distribution of reinforced beams along the cylindrical shell.

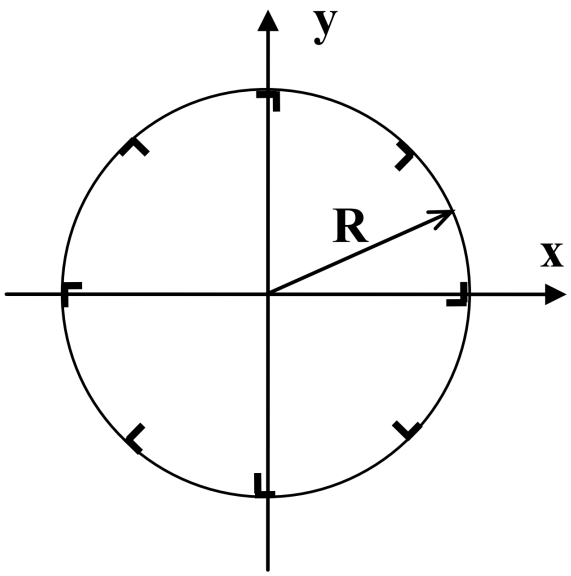


Figure 4: Circular symmetrically distributed L beams.

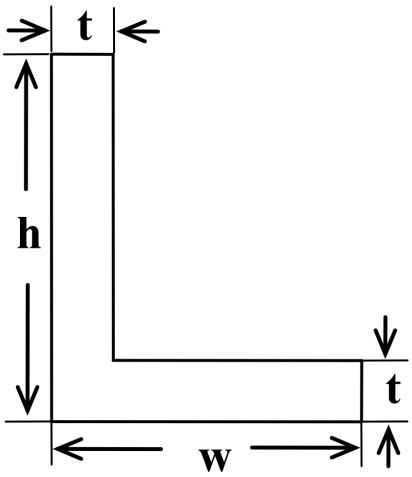


Figure 5: Cross-section of L beam.

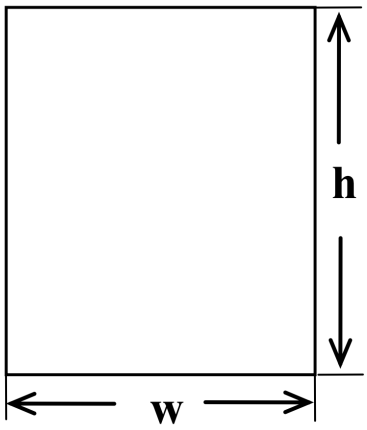


Figure 6: Cross-section of rectangle beam.

1.2 Material Parameters

The material properties used in the baseline model are as follows:

Copper $E = 100\,\text{GPa}$, $\nu = 0.33$.

Composite Facesheet: $E_{11} = 101\,\text{GPa}$, $E_{22} = 10.1\,\text{GPa}$, $G_{12} = 15\,\text{GPa}$, $\nu_{12} = 0.1$.

Composite Core: $E_{11} = E_{22} = 100\,\text{MPa}$, $G_{12} = 50\,\text{MPa}$, $G_{13} = G_{23} = 1.06\,\text{MPa}$, $\nu_{12} = 0.3$.

1.3 Load Specification

The internal surface distributed pressure follows a spatially varying function:

$$P(y) = 200|y| \quad [\mathrm{N/m^2}] \tag{1}$$

where y is the vertical coordinate (Figure 2). This creates a maximum pressure of 100 Pa at the top and bottom of the cylinder ($y = \pm R$) and zero pressure at the sides ($y = 0$).

1.4 Composite Shell Configuration

The baseline laminate consists of 5 layers with the following configuration:

- **Layer 1:** Facesheet, 0.3 mm, $+45°$
- **Layer 2:** Facesheet, 0.3 mm, $-45°$
- **Layer 3:** Core, 5.0 mm, $0°$
- **Layer 4:** Facesheet, 0.3 mm, $-45°$
- **Layer 5:** Facesheet, 0.3 mm, $+45°$

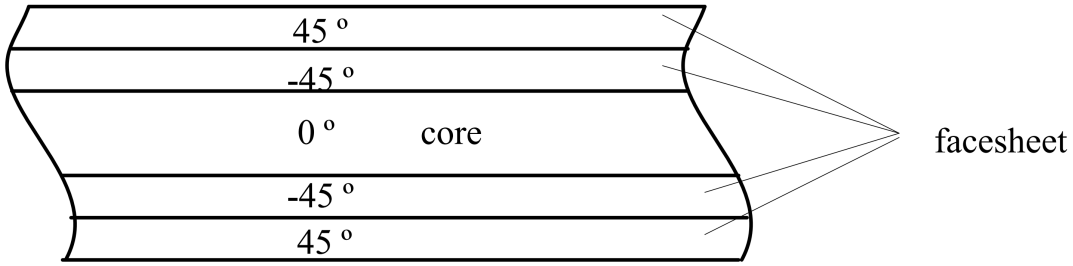


Figure 7: Layers of the composite material.

1.5 Structure Construction

Phase 1: Preparation & Unit Consistency

MSC.Nastran is a unitless solver. A mismatch between geometry units (mm) and material stiffness (Pascals or $\mathrm{N/m^2}$) will cause the structure to fail.

- Note: All the data must be in Meters (m), Newtons (N), and Pascals ($\mathrm{N/m^2}$).
- Open Patran, create a new database (cylinder.db).
- Set Analysis Code to MSC.Nstran and Analysis Type to Structural.

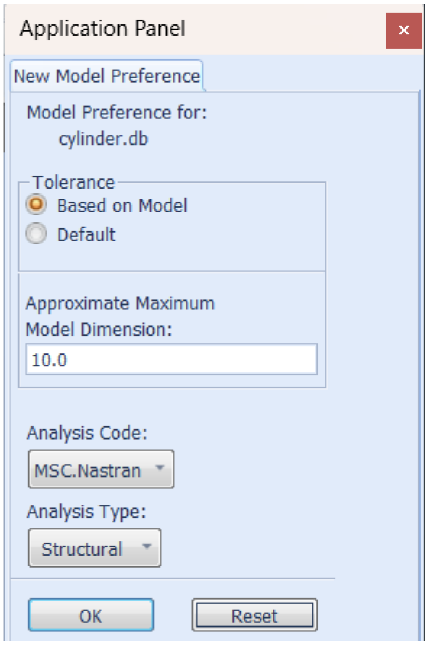


Figure 8: Initial preferences panel.

Phase 2: Geometry Creation

1. Create the Base Points:

- Go to Geometry → Action: Create → Object: Point → Method: XYZ.
- Enter [0.5 0 0] and click Apply.
- Change Action: Transform → Object: Point → Method: Rotate.
- Enter Rotation Angle: 45, Repeat Count: 7, Point List: [0.5 0 0].
- Select the point you just created and click Apply.

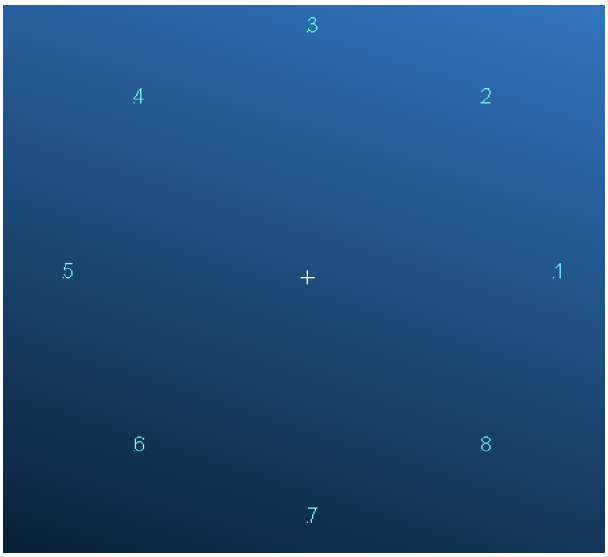


Figure 9: Points creation.

2. Create and Break the Base Circle:

- Change Action: Create → Object: Curve → Method: 2D Circle.
- Radius: 0.5, Center: [0 0 0]. Click Apply.

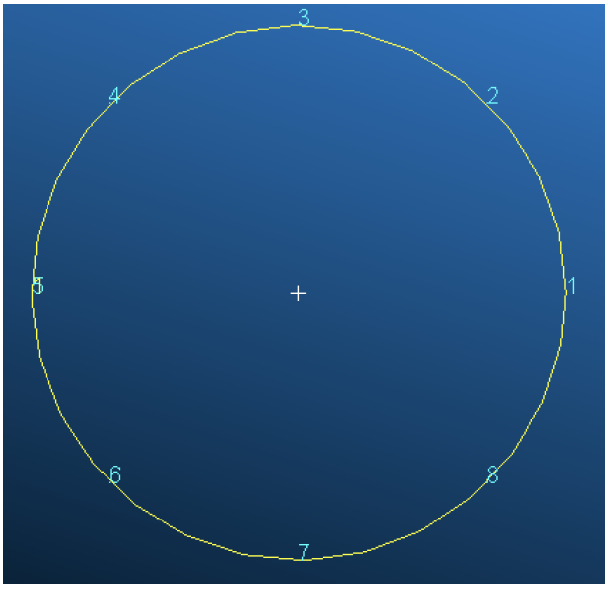


Figure 10: Circular curve creation.

- Change Action: Edit → Object: Curve → Method: Break.
- Change Option to Point.
- Select the circle, then select the 8 points. Click Apply.
- Note: When asked to delete the original curve, click YES.

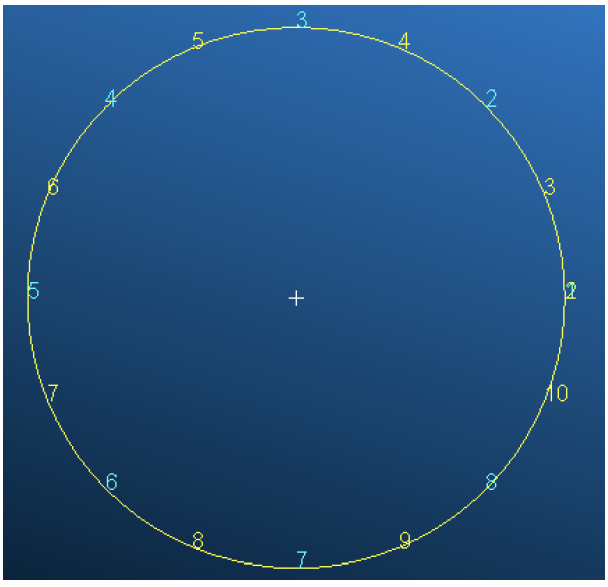


Figure 11: Circular curve breaking.

3. Extrude the Composite Half (Z=0 to Z=3):

- Change Action: Create → Object: Surface → Method: Extrude.
- Translation Vector: <0 0 3>.
- Select the 8 arcs. Click Apply.

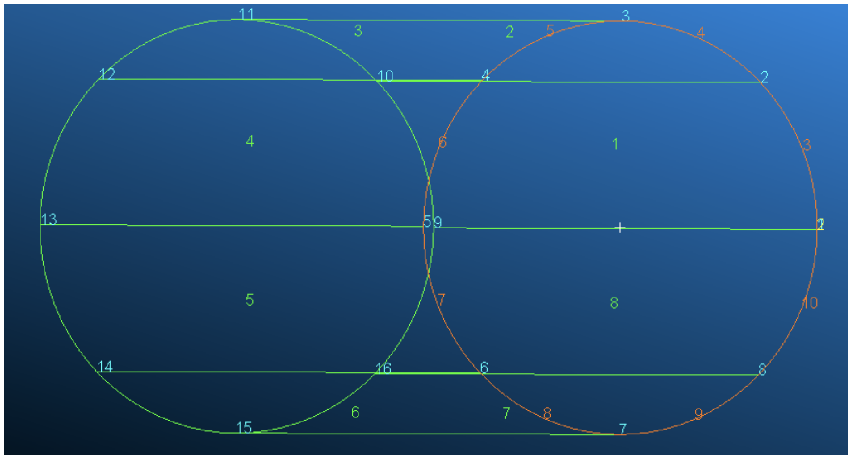


Figure 12: First section extrusion.

4. Extrude the Metal Half (Z=3 to Z=6):

- Keep the same Extrude settings (<0 0 3>).
- Select the 8 top edges of the surfaces you just created (the arcs at Z=3).
- Click Apply. The geometry is now flawlessly connected end-to-end.

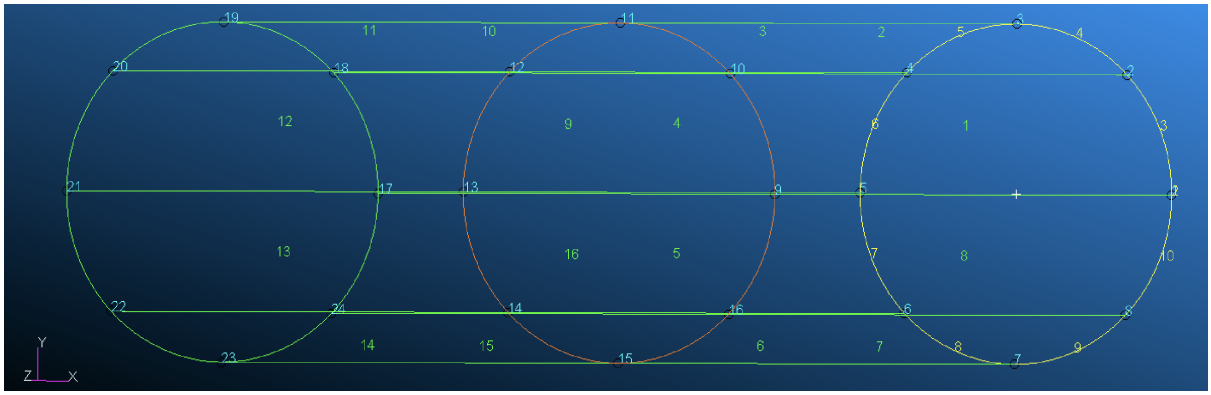


Figure 13: Second section extrusion.

Phase 3: Material and Laminate Setup

- **Copper:** Create → Isotropic → Manual Input. Name: Copper. Input Properties: Elastic Modulus = 1.0e11, Poisson Ratio = 0.33.

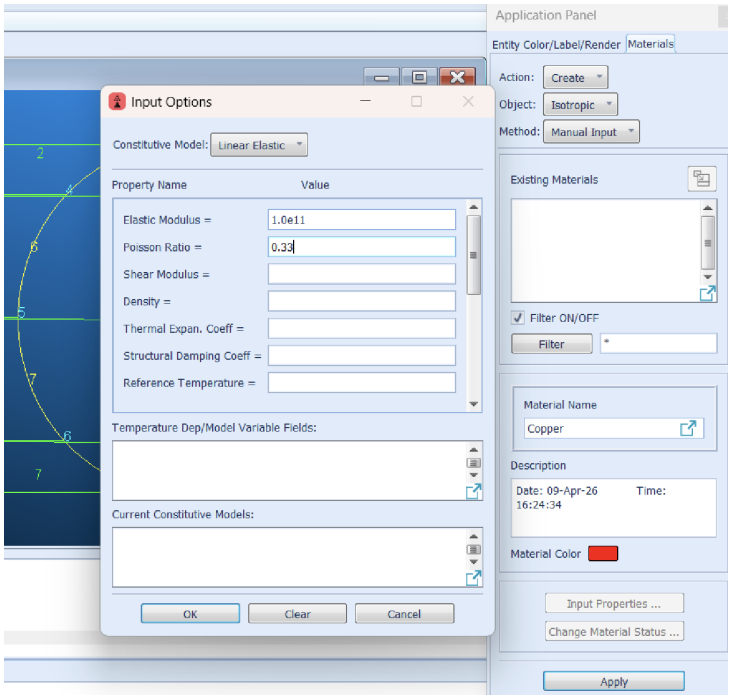


Figure 14: Copper material creation.

- **Composite Constituents:** Create → 2D Orthotropic. Name: Facesheet. Input Properties: E11= 1.0e11, E22= 1.0e10, Poisson Ratio 12= 0.1, Shear Modulus 12= 1.50e10.
- **Laminate Layup:** Create → Composite → Laminate. Name: Layup. Build the table from top to bottom with the appropriate layers and angles.

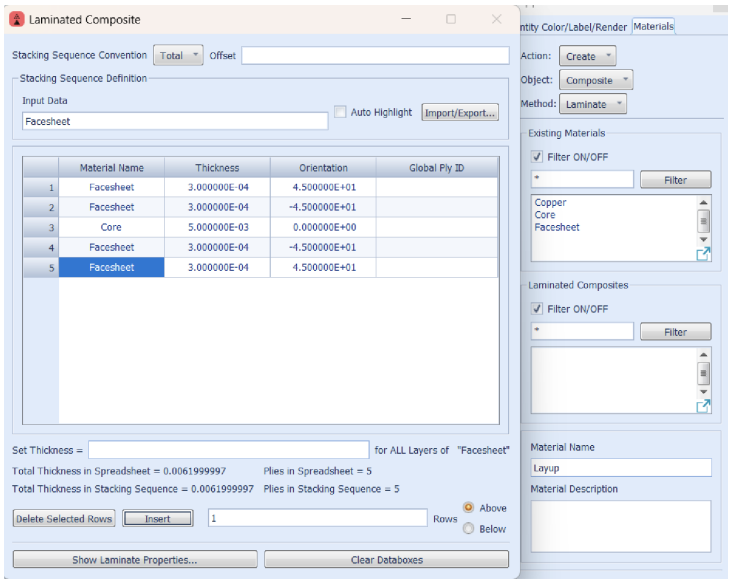


Figure 15: Shell material creation.

Phase 4: Element Properties

1. **Longitudinal L-Beams:**

- Create → 1D → Beam. Name: L_Beams.
- Input Properties: Material: Copper.
- Click Beam Library: Select L-Shape. Input dimensions in meters: W= 0.01, H= 0.01, t1= 0.003, t2= 0.003.

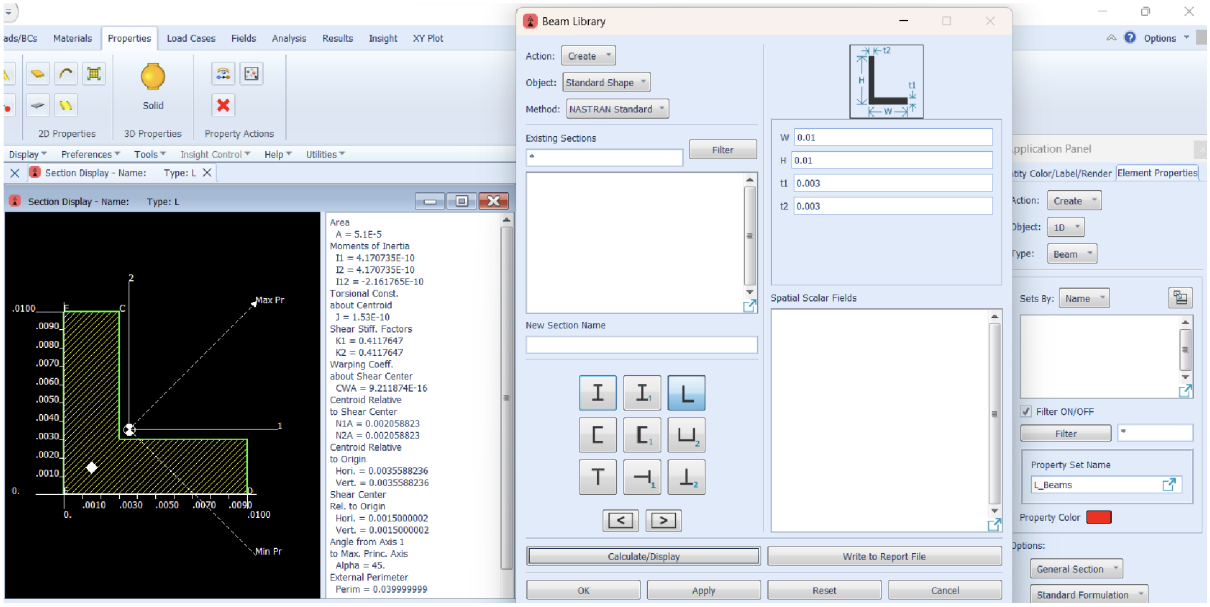


Figure 16: L-Beam cross-sectional determination.

- Set Bar Orientation Vector to <1 0 0>.

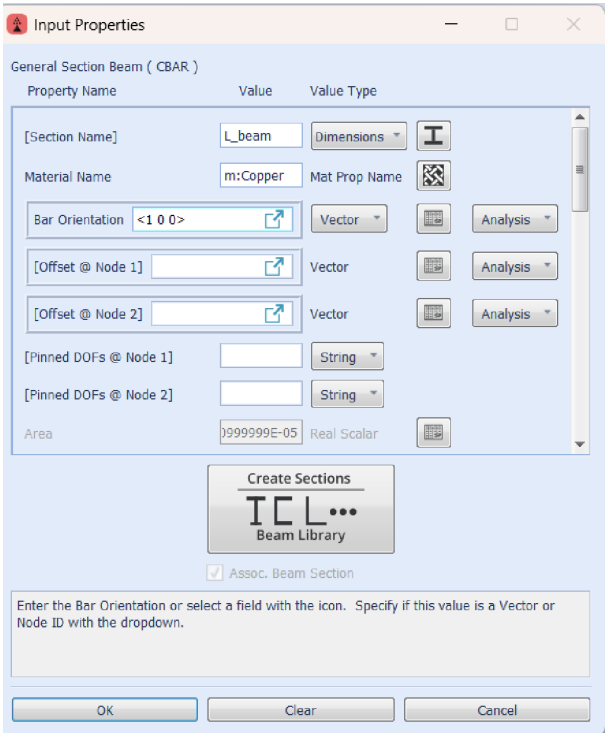


Figure 17: L-Beam creation.

- Application Region: Select the 16 longitudinal curves (8 bottom, 8 top). Click Apply.

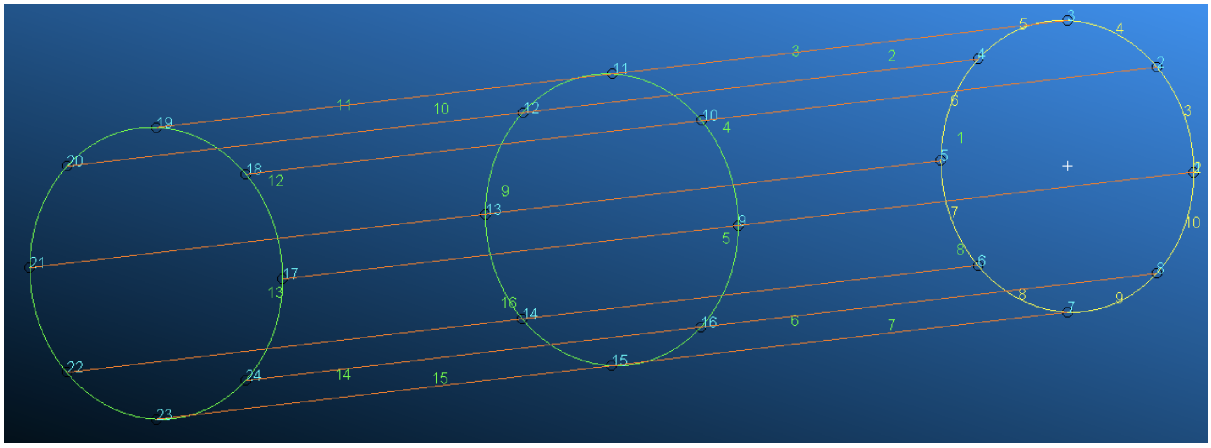


Figure 18: Sections that will have the L-beam cross sectional shape.

2. **Circular Rectangular Beams:**

   - Create → 1D → Beam. Name: Rect_Beams.
   - Input Properties: Material: Copper.
   - Click Beam Library: Select Solid Rectangular. Input: W = 0.01, H = 0.01.
   - Set Bar Orientation Vector to $\langle 0, 0, 1 \rangle$.
   - Application Region: Select the 24 circular arcs (8 at Z=0, 8 at Z=3, 8 at Z=6). Click Apply.

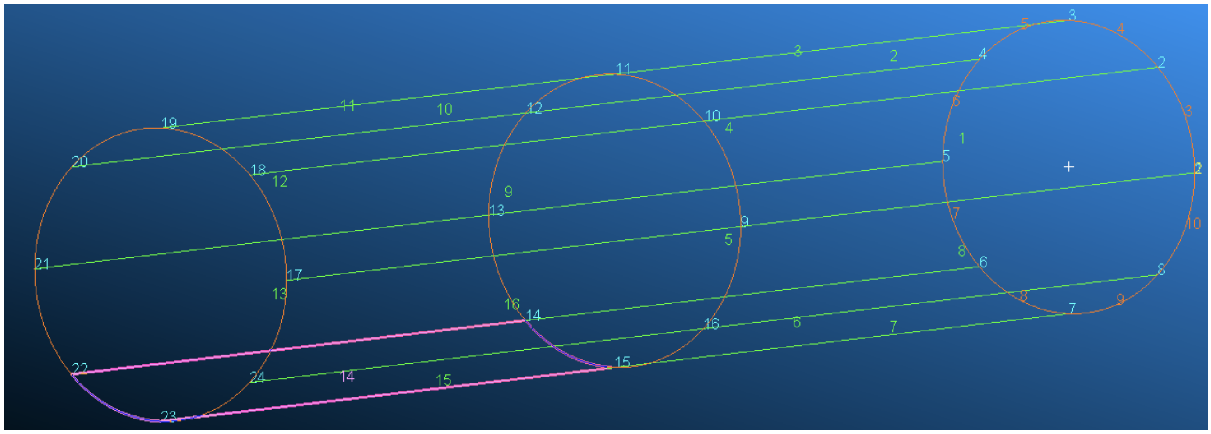


Figure 19: Sections that will have the rectangle cross sectional shape.

3. **Metal Shell:**

   - Create → 2D → Shell. Name: Metal_Shell.
   - Input Properties: Material: Copper. Thickness: 0.002.
   - Application Region: Select the 8 top surfaces (Z=3 to Z=6). Click Apply.

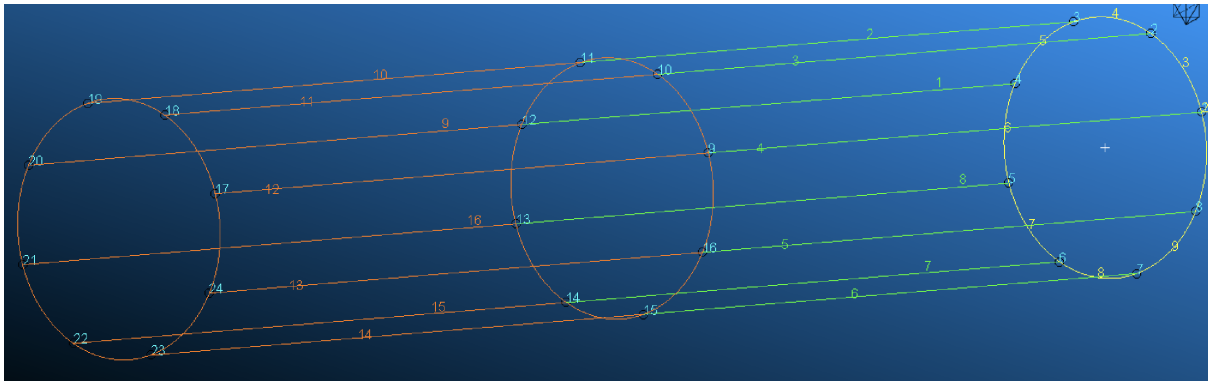


Figure 20: Sections that will have the Metal Shell material.

4. **Composite Shell:**

   - Create → 2D → Shell. Name: Comp_Shell.
   - Input Properties: Material: Layup. Thickness: 0.0062.
   - Application Region: Select the 8 bottom surfaces (Z=0 to Z=3). Click Apply.

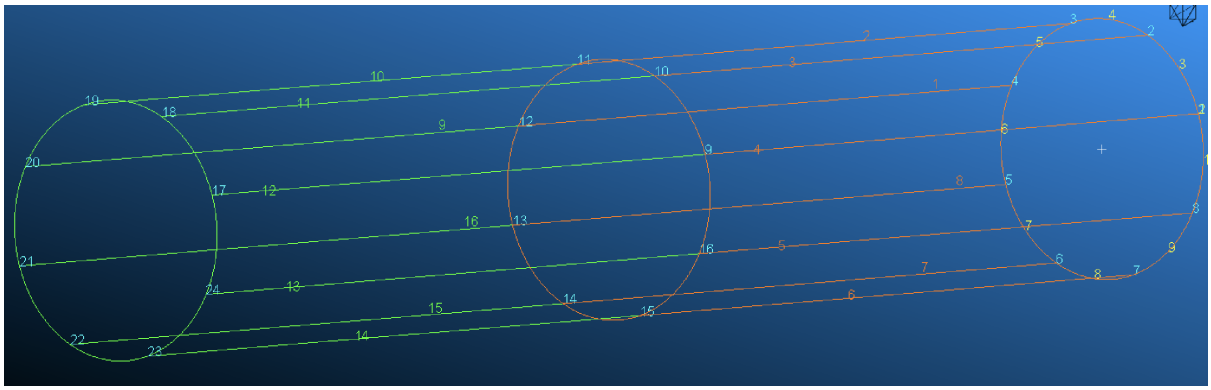


Figure 21: Sections that will have the Composite Shell material.

Phase 5: Meshing & Equivalencing

1. Mesh the surfaces:

   - Create → Mesh → Surface. Topology: Quad4.
   - Global Edge Length: 0.05. Select all 16 surfaces. Click Apply.

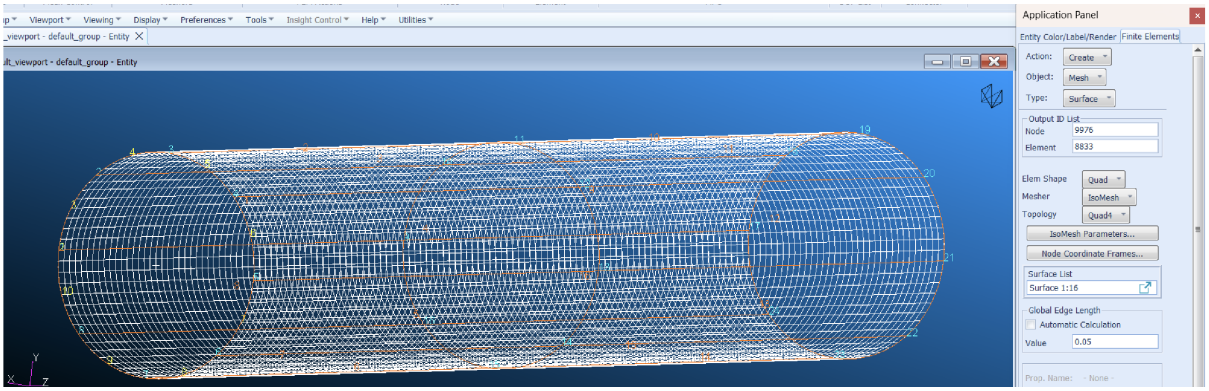


Figure 22: Surface meshing.

2. Mesh the curves:

   - Create → Mesh → Curve. Topology: Bar2
   - Global Edge Length: 0.05. (This MUST exactly match the surface edge length). Select all 16 longitudinal curves and 24 circular curves. Click Apply.

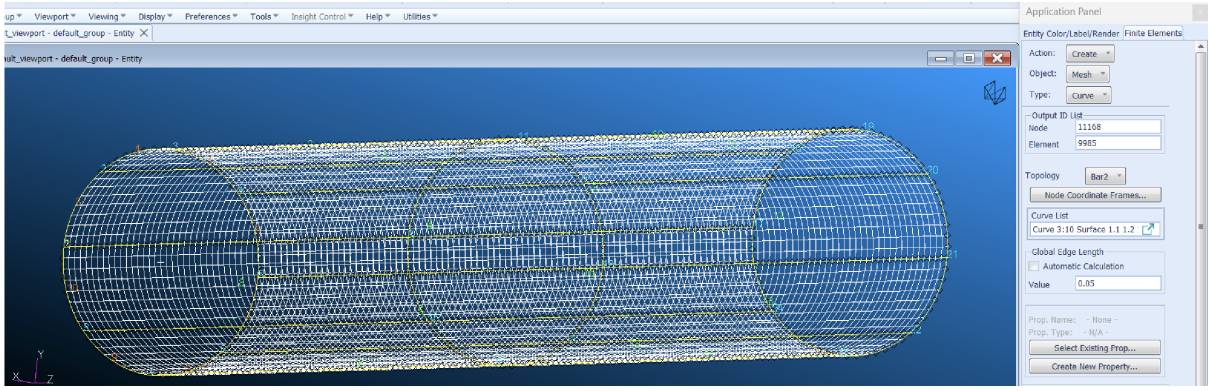


Figure 23: Curves meshing.

3. Equivalence:

   - Action: Equivalence → Object: All → Method: Tolerance Cube.
   - Tolerance: 0.005. Click Apply.

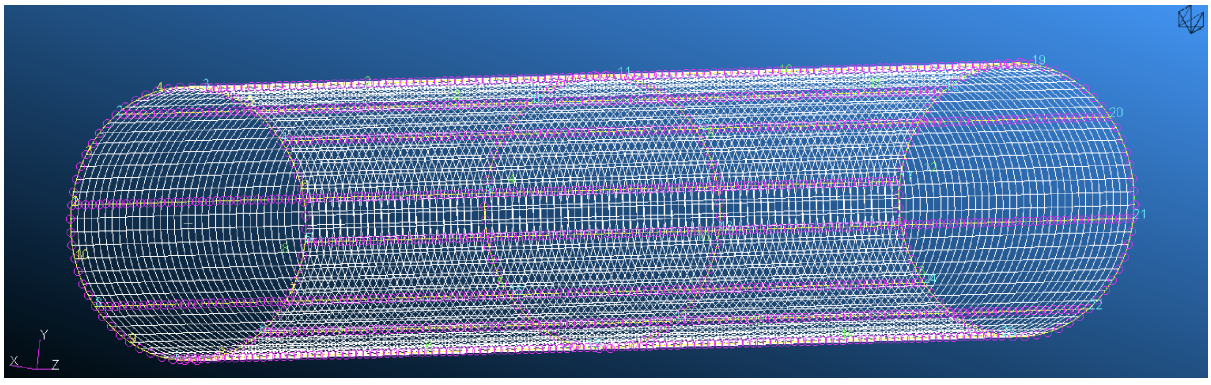


Figure 24: Equivalence result.

Phase 6: Loads and Boundary Conditions

1. Spatial Field:
   - Go to Fields tab.
   - Create → Spatial → PCL Function. Name: Press_Field.
   - Scalar Function: 200 * abs('Y). Click Apply.
2. "Fixed" Constraint:
   - Go to Loads/BCs tab.
   - Create → Displacement → Nodal. Name: Fixed_End.
   - Input Data: Translations: <0 0 0>, Rotations: <0 0 0>.
   - Application Region: Select the 8 bottom arcs at Z=0. Click Apply.

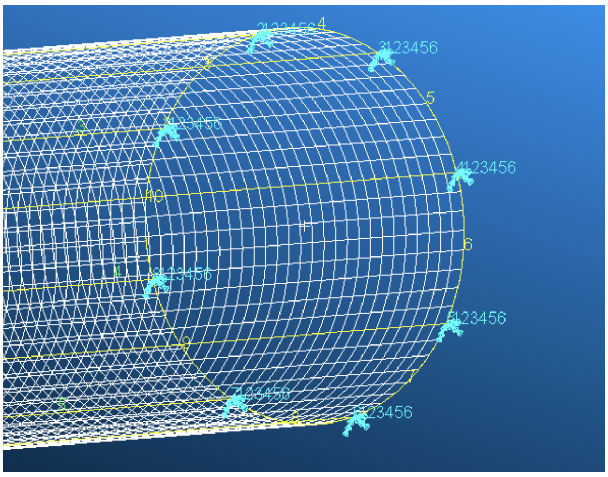
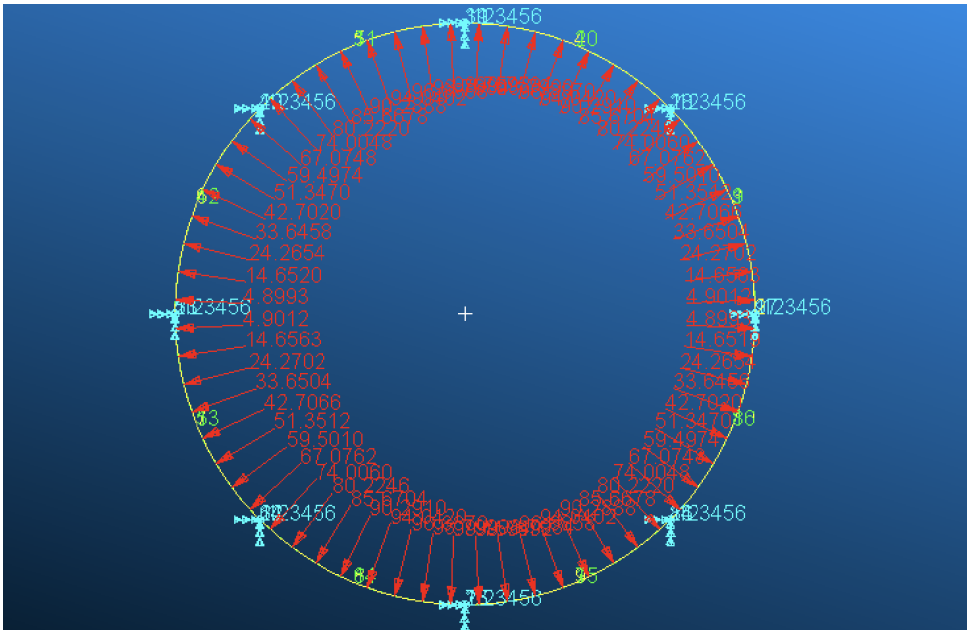


Figure 25: Visualization of the fixed end of the structure.

3. Pressure Load Visualization:
   - Create → Pressure → Element Uniform. Target: 2D. Name: Pressure.
   - Input Data: Click the spatial field icon and select Press_Field.
   - Application Region: Change the geometry filter to FEM. Drag a box over all 2D elements. Click Add, OK, Apply. (Red arrows will fill the inside of the cylinder, pointing outward).


123456
123456
123456

Phase 8: Post-Processing & Animation

- In Analysis tab, change Action: Access Results, Object: Attach XDB. Select the created file and click Apply.
- View Stresses: Go to Results → Quick Plot. Select Stress Tensor → von Mises and Displacements, Translational.
- Fix Animation Scaling: Click the Display Attributes icon, then change Scale Interpretation from Model Scale to True Scale, and set Scale Factor to 1.0. Click Apply.

2. Analysis Results

2.1 Displacement Response

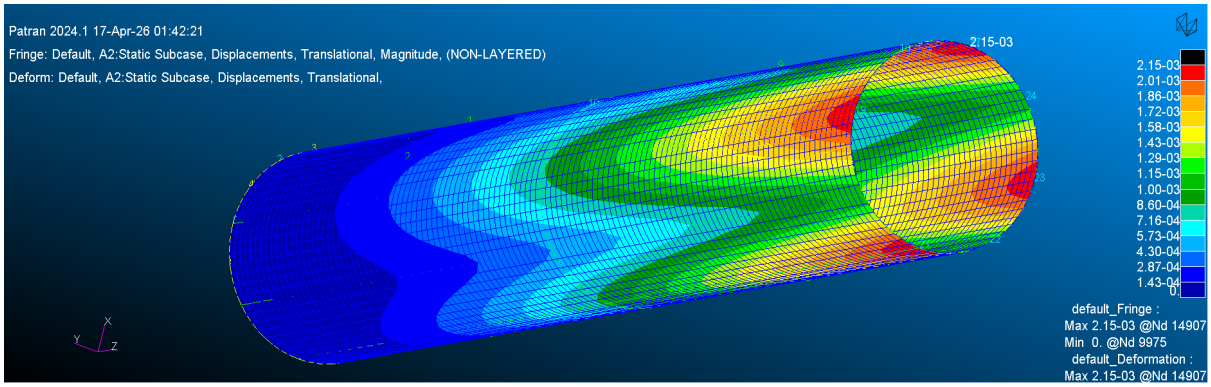


Figure 28: Translational displacement magnitude of the cylindrical shell.

2.2 Stress Distribution

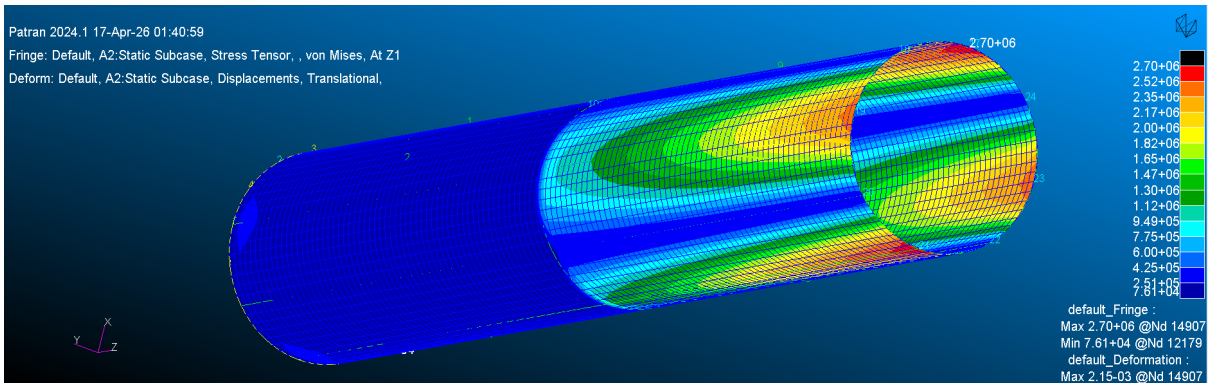


Figure 29: Von Mises stress distribution in the cylindrical shell.

2.3 Results Discussion

Displacement Response:

The translational displacement of the cylindrical shell is shown in figure 28. The displacement

is greatest at the free end of the shell, with a clear variation across the length of the structure. The displacement reaches a maximum of $\delta_{\max} = 2.15\,\text{mm}$, which is expected given the applied internal pressure. The displacement also exhibits a tendency to bias toward the top and bottom of the shell (and right and left subsequently), where the internal pressure is highest due to the pressure distribution function. This variation is a direct result of the pressure load and the geometrical configuration of the shell.

Stress Distribution:
The Von Mises stress distribution across the cylindrical shell is shown in figure 29. The stress concentration is observed around the free end of the shell, with a peak of $\sigma_{\max} = 27\,\text{MPa}$. Additionally, the stress distribution follows the load pattern, with higher stress observed at regions subjected to higher pressure, particularly around the top and bottom of the shell (and right and left subsequently). The smooth stress distribution observed in other parts of the shell suggests that the material and geometry of the shell are performing well in terms of distributing the load efficiently, while the localized concentrations at specific areas highlight the importance of further optimization in these high-stress zones.

Both results provide crucial insights into the structural performance of the cylindrical shell under the given load conditions and suggest areas where material properties or structural modifications could be explored for improved performance.

3. Structural Optimization

3.1 Objectives and Constraints

The optimization campaign aims to minimize structural mass while satisfying multi-physics constraints:

Constraint Type	Limit	Justification
Maximum Stress (Al)	$\sigma_{\text{vm}} \leq 200\,\text{MPa}$	1.37× safety factor on yield
Maximum Displacement	$\delta_{\text{tip}} \leq 5\,\text{mm}$	$L/1200$ deflection limit for precision alignment
Buckling Factor	$\lambda \geq 2.0$	Eigenvalue stability requirement
Mass Objective	$\min(M_{\text{total}})$	Primary objective function

Table 1: Optimization Constraints and Targets

3.2 Design Parameters

3.2.1 Material Selection

The selection of Aluminum 6061-T6 over the baseline copper and alternative titanium alloys represents a multi-objective decision balancing specific stiffness, manufacturability, and cost.

Material	E [GPa]	ρ [kg/m^3]	σ_y [MPa]	**Specific Stiffness**	**R. Cost**	**Weldability**
Copper	100	8,960	70–220	12.3 [10^6 m^2/s^2]	1.0	Excellent
Al 6061-T6	69	2,700	276	**25.6** [10^6 m^2/s^2]	0.4	Excellent
Ti-6Al-4V	110	4,500	880	24.4 [10^6 m^2/s^2]	4.0	Poor

Table 2: Comparative Material Analysis for Metallic Components.

Analysis:

While copper offers higher absolute stiffness ($E = 110\,\text{GPa}$), its density ($\rho = 8960\,\text{kg/m}^3$) results in poor specific stiffness (12.3×10^6). Titanium provides superior strength-to-weight ratio but at 4× material cost and poor weldability, requiring expensive friction stir welding for the ring frame attachments.

As such, aluminum 6061-T6 provides the optimal specific stiffness (25.6×10^6), nearly double that of copper and exceeding titanium. This allows the cantilever structure to minimize mass while maintaining deflection limits governed by EI. The excellent weldability enables conventional MIG/TIG joining of the L-beams to the shell, whereas titanium would require mechanical fastening (rivets), adding parasitic mass and stress concentrations.

3.2.2 Composite Angle Optimization: [90/0] vs. [±45]

The laminate orientation significantly affects structural efficiency. Using Classical Lamination Theory (CLT), the extensional and bending stiffness matrices are evaluated as

$$[A] = \sum_{k=1}^{n} \bar{Q}_k \left(z_k - z_{k-1}\right) \tag{2}$$

$$[D] = \frac{1}{3} \sum_{k=1}^{n} \bar{Q}_k \left(z_k^3 - z_{k-1}^3\right) \tag{3}$$

where $\bar{Q}_k$ is the transformed reduced stiffness matrix of the kth ply, and z_k are the laminate interface coordinates measured from the mid-plane.

Using the stated material properties and ply thicknesses, the baseline laminate [45/ − 45/Core/ − 45/45] and the optimized laminate [90/0/Core/0/90] give the following stiffnesses:

Property	[45/ − 45/Core/ − 45/45]	[90/0/Core/0/90]
A_{11} (Axial) [MN/m]	52.52	**67.28** (+28%)
A_{22} (Hoop) [MN/m]	52.52	**67.28** (+28%)
A_{66} (In-plane shear) [MN/m]	**33.01**	18.25 (-45%)
D_{11} (Axial bending) [N m]	410.15	**480.42** (+17%)
D_{22} (Hoop bending) [N m]	410.15	**572.14** (+39%)

Table 3: Laminate Stiffness Comparison (per unit width).

Physical Interpretation:

1. **Higher normal stiffness in the principal directions:** The [90/0/Core/0/90] laminate increases both A_{11} and A_{22} by about 28%, meaning that it is stiffer in the axial and circumferential directions. Since the shell response is mainly governed by normal stresses in these two directions together with cantilever bending, this stacking sequence is mechanically more suitable than the angle-ply baseline.

2. **Improved bending stiffness with a shear trade-off:** The optimized laminate increases D_{11} by about 17% and D_{22} by about 39%, which improves resistance to bending deformation. However, this benefit comes with a reduction in in-plane shear stiffness, since A_{66} decreases by about 45%. This means the cross-ply laminate is better for axial and hoop stiffness, while the [±45] laminate remains better for shear-dominated loading.

3. **Reduction of coupling effects:** The baseline [45/ − 45/Core/ − 45/45] laminate has $A_{16} = A_{26} = 0$, so it does not exhibit extensional-shear coupling. However, it retains nonzero bending-twisting coupling terms ($D_{16}, D_{26} \neq 0$), which can produce unwanted warping under bending. The cross-ply [90/0/Core/0/90] laminate eliminates these coupling terms, leading to a cleaner and more predictable structural response.

Failure Mode Analysis:

Under the representative combined loading of hoop force $N_\theta = 50\,\mathrm{N/m}$ and bending moment $M_x = 400\,\mathrm{Nm/m}$, the two laminate configurations show different governing failure mechanisms:

For the [±45] **laminate:** the axial and circumferential loads are transferred mainly into in-plane shear in the ply coordinates. As a result, the critical response is matrix-dominated shear, which makes the laminate more vulnerable to shear-driven damage.

For the [90/0] **laminate:** the same loading is carried more directly as fiber-direction normal stress. The 90° plies support the hoop response, while the 0° plies carry the axial and bending loads. This produces a more efficient load path and a more favorable failure mode.

Therefore, the optimized cross-ply laminate is structurally better suited for this shell because it reduces shear-dominated damage and uses the fiber stiffness more effectively under combined membrane and bending loading.

3.3 Optimization Results

Three cases were analyzed during this optimization process:

1. 2mm Aluminum sheet thickness.

2. 3mm Aluminum sheet thickness.

3. 3mm Aluminum sheet thickness and [90/0] Composite layer configuration.

1. 2mm Aluminum sheet thickness

Displacement Response

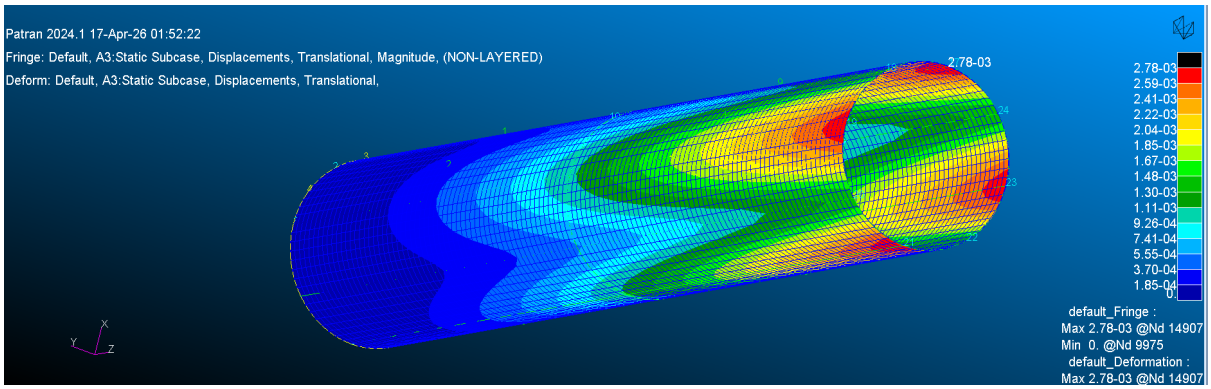


Figure 30: Translational displacement magnitude of the cylindrical shell with 2mm Aluminum sheet thickness.

Stress Distribution

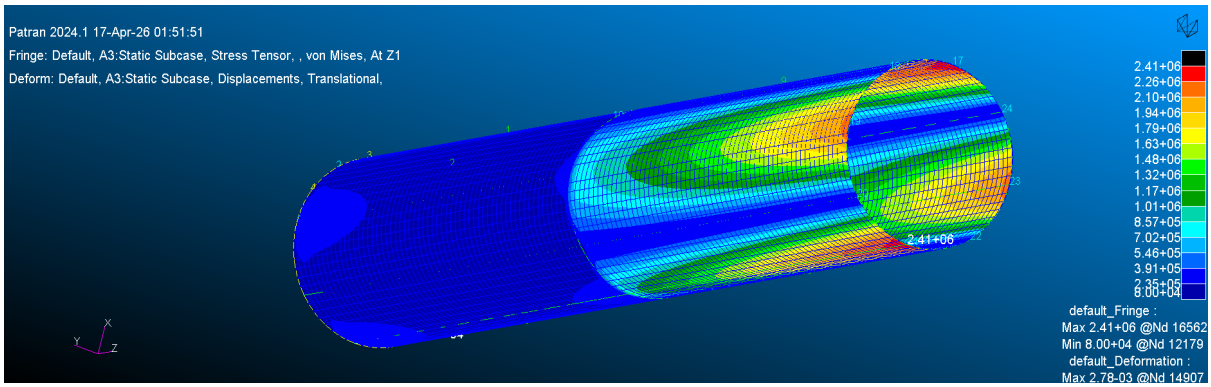


Figure 31: Von Mises stress distribution in the cylindrical shell with 2mm Aluminum sheet thickness.

2. 3mm Aluminum sheet thickness

Displacement Response

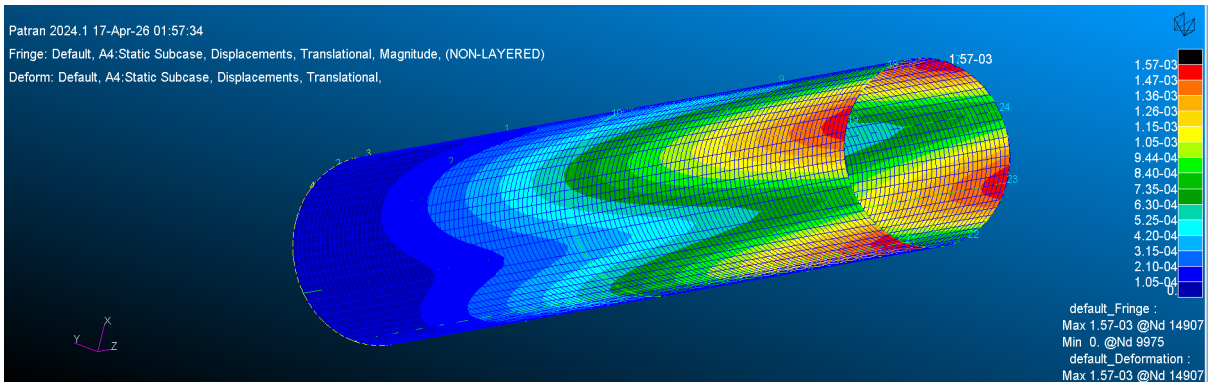


Figure 32: Translational displacement magnitude of the cylindrical shell with 3mm Aluminum sheet thickness.

Stress Distribution

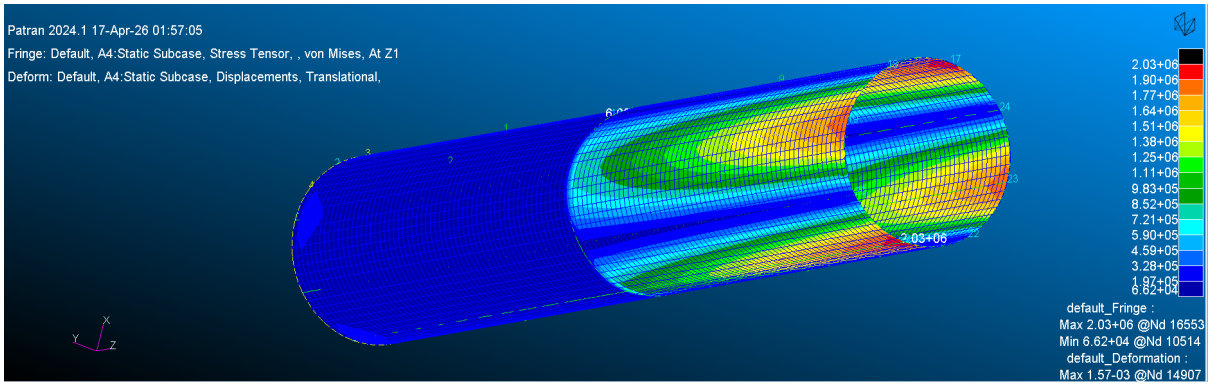


Figure 33: Von Mises stress distribution in the cylindrical shell with 3mm Aluminum sheet thickness.

3. 3mm Aluminum sheet thickness and [90/0] Composite layer configuration

Displacement Response

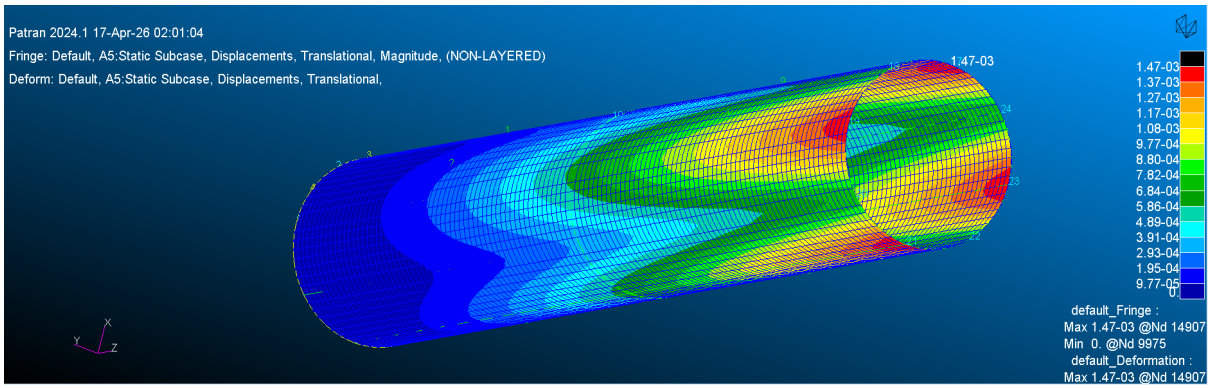


Figure 34: Translational displacement magnitude of the cylindrical shell with 3mm Aluminum sheet thickness and [90/0] Composite layerconfiguration.

Stress Distribution

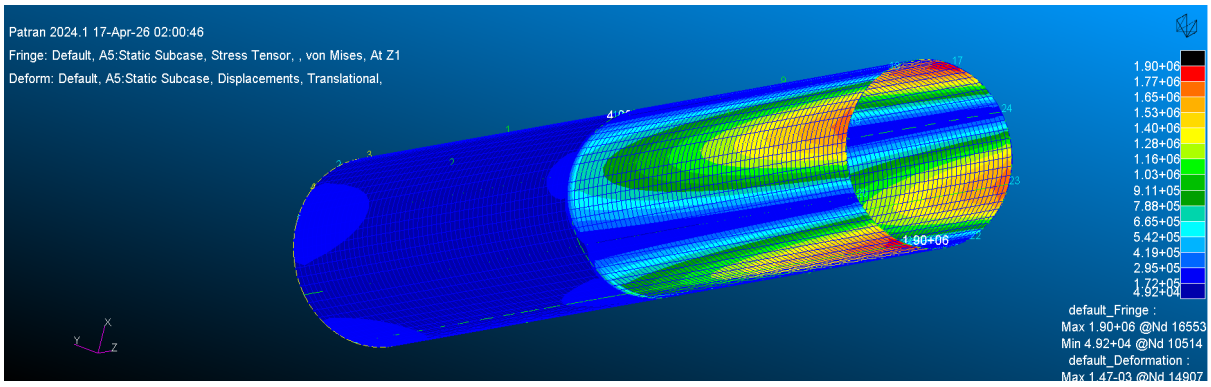


Figure 35: Von Mises stress distribution in the cylindrical shell with 3mm Aluminum sheet thickness and [90/0] composite configuration.

3.4 Optimization Results Discussion

The optimized design achieves substantial mass reduction while improving structural performance:

Parameter	Baseline	Optimized	Improvement
Total Mass	100% (reference)	31%	-69%
Max Displacement	2.15 mm	1.47 mm	-31.63%
Max von Mises Stress	27 MPa	19 MPa	-29.63%
Safety Factor	8.15	14.53	+78%
Cost	100% (reference)	approx 40%	approx -60%

Table 4: Baseline vs. Optimized Design Comparison.

4. Problems Encountered and Solutions

Several challenges arose during the project execution, ranging from theoretical understanding to software-specific technical issues:

Conceptual Understanding: Initial difficulties in comprehending the coupling between membrane and bending behavior in composite shells required review of Classical Lamination Theory (CLT) and shear deformation theory. Resources including "Mechanics of Composite Materials" by Robert M. Jones and MIT OpenCourseWare lectures on anisotropic elasticity proved essential for understanding why the [90/0] layup outperforms angle-ply configurations under pressure loading.

Software Interface Familiarization: Mastering the MSC.Patran interface presented a steep learning curve, particularly regarding the geometric modeling workflow and the distinction between geometric entities (curves/surfaces) and finite element entities (nodes/elements). Video tutorials from MSC's official learning center and academic YouTube channels demonstrating Patran-Nastran workflows for pressure vessel analysis were instrumental in understanding the proper sequence of geometry creation → meshing → equivalence → loading.

File Extension and Format Compatibility: A persistent issue involved the output format selection between XDB (Patran binary) and OP2 (Nastran binary). Initial attempts using OP2 format resulted in unreadable results files in Patran 2024. The solution required explicitly setting the Translation Parameters to XDB format in the Analysis → Solution Type menu before submission. Additionally, ensuring the database path contained no spaces or special characters prevented read/write errors during large solution files.

5. Opinions and Suggestions

Educational Value: This finite element analysis project proved to be an exceptionally informative and practical educational experience, bridging the gap between theoretical solid mechanics and industrial engineering practice. The comprehensive nature of the cylindrical shell problem—encompassing composite material modeling, spatially varying pressure loads, multi-material interfaces, and optimization constraints—provided a holistic understanding of structural analysis that textbook problems rarely capture. The in-class examples completed under the direct guidance of Prof. Shenyan CHEN were particularly valuable; these step-by-step demonstrations of Patran-Nastran workflows, especially regarding the critical nuances of unit consistency and geometric equivalencing, prevented hours of frustration and established a logical framework for approaching complex simulations. The iterative process of troubleshooting mesh connectivity issues and interpreting stress concentrations developed an engineering intuition for finite element modeling that will be directly applicable to future aerospace structural design challenges.

Curriculum Recommendations: While the course provided an excellent foundation, expanding the duration to include dedicated sessions on solver theory and results interpretation would significantly enhance the learning outcomes. Specifically, additional lectures explaining the mathematical distinctions between different Nastran solution sequences—such as SOL 101 (Linear Static), SOL 103 (Modal), SOL 105 (Buckling), and SOL 106 (Nonlinear)—would help students understand not just how to click through menus, but which physical phenomena each solver captures and when linear assumptions break down. Furthermore, structured sessions dedicated to interpreting "bad" versus "good" results—teaching students to identify mesh refinement issues, hourglass modes, stress singularities at constraints, and checkerboarding in topology optimization—would develop critical judgment skills. Such sessions could include deliberate "broken" models for students to diagnose, fostering the debugging mindset essential for professional FEA analysts.